

Channel-Aware Laplace Approximation for the Posterior over p

This note collects the derivations for a channel-aware Laplace approximation of the posterior over the macroscopic occupancy vector p of an ensemble of Markov channels, based on a single interval-averaged current measurement.

The main goals are:

- to obtain an explicit formula for the *Gaussian (Laplace) posterior covariance* Σ_{post} in terms of the prior covariance Σ_p and a rank-1 update defined by the measurement;
- to write the *mean update* in a form that does not involve any explicit inverse of the (singular) prior covariance;
- to emphasise that the posterior mean used in practice is the *MAP estimate* obtained by Newton iteration, and the covariance is the *Laplace covariance* given by the curvature at the MAP.

Throughout, we keep the dependence of all intermediate objects on p explicit: $\delta(p)$, $V(p)$, $v(p)$, etc. Whenever we drop the argument and write δ , V , v , they are implicitly evaluated at the current expansion point (either a fixed p_0 , or the current iterate $p^{(k)}$ in Newton's method).

Coordinates, simplex and singular covariance. The macroscopic occupancy vector p lives on the simplex $\{p \in \mathbb{R}^{1 \times d} : p_i \geq 0, \sum_i p_i = 1\}$, a $(d-1)$ -dimensional manifold embedded in $\mathbb{R}^{1 \times d}$. The prior covariance Σ_p is therefore singular as a $d \times d$ matrix in ambient coordinates. Formally, one can view Σ_p^{-1} as the inverse on the $(d-1)$ -dimensional tangent subspace or as a pseudo-inverse, but *in all final expressions we use in implementation, Σ_p^{-1} cancels algebraically*. We never need to numerically invert a singular covariance.

1. Setup and Notation

We consider:

- A macroscopic occupancy vector

$$p \in \mathbb{R}^{1 \times d},$$

a row vector of state probabilities.

- A per-state conductance vector

$$\gamma \in \mathbb{R}^{d \times 1},$$

a column vector.

- A per-state intrinsic-variance vector

$$\sigma^2 \in \mathbb{R}^{d \times 1}.$$

The prior over p (restricted to the simplex) is Gaussian:

$$p \sim \mathcal{N}(\mu, \Sigma_p),$$

where $\mu \in \mathbb{R}^{1 \times d}$ is the prior mean and $\Sigma_p \in \mathbb{R}^{d \times d}$ is the prior covariance. As noted above, Σ_p is singular in the ambient coordinates.

We observe a scalar interval-averaged current y :

$$y \approx N_{\text{ch}}(p \cdot \gamma) + \text{noise},$$

with N_{ch} channels and:

- extrinsic (measurement) noise variance ϵ^2 ;
- intrinsic ensemble variance $N_{\text{ch}}(p \cdot \sigma^2)$ arising from channel stochasticity.

Given a candidate state p , define

$$\begin{aligned} \delta(p) &= y - N_{\text{ch}}(p \cdot \gamma), \\ V(p) &= \epsilon^2 + N_{\text{ch}}(p \cdot \sigma^2), \end{aligned}$$

and the *tilted* vector

$$v(p) = \gamma + \frac{\delta(p)}{V(p)} \sigma^2 \in \mathbb{R}^{d \times 1}.$$

We work with an energy function $E(p)$ proportional to the negative log-posterior,

$$p(p \mid y) \propto \exp\left(-\frac{1}{2}E(p)\right).$$

We will denote:

- $\mathbf{g}(p) = \nabla E(p)$, a *row* vector in $\mathbb{R}^{1 \times d}$;
- $\mathbf{H}(p) = \nabla^2 E(p)$, a $d \times d$ Hessian.

2. Laplace Approximation Around an Expansion Point

We approximate the posterior around an expansion point p_0 via a second-order Taylor expansion:

$$E(p) \approx E(p_0) + (p - p_0) \mathbf{g}_0^\top + \frac{1}{2}(p - p_0) \mathbf{H}_0 (p - p_0)^\top,$$

with

$$\mathbf{g}_0 = \nabla E(p_0), \quad \mathbf{H}_0 = \nabla^2 E(p)|_{p=p_0}.$$

Plugging this into

$$p(p \mid y) \propto \exp\left(-\frac{1}{2}E(p)\right)$$

gives a Gaussian approximation

$$p(p \mid y) \approx \mathcal{N}(p_{\text{post}}, \Sigma_{\text{post}}),$$

with

$$\Sigma_{\text{post}} \approx 2 \mathbf{H}_0^{-1},$$

$$p_{\text{post}} \approx p_0 - \mathbf{g}_0 \mathbf{H}_0^{-1} = p_0 - \frac{1}{2} \mathbf{g}_0 \Sigma_{\text{post}}.$$

Conceptually, the Laplace covariance is the inverse curvature at the chosen expansion point, and the mean is the Newton step from p_0 . In practice, we will:

- use a special rank-1 structure of $\mathbf{H}_0$ to obtain a closed-form Σ_{post} in terms of Σ_p ;
- eliminate any explicit Σ_p^{-1} from the mean update;
- iterate the Newton step until convergence to obtain the MAP.

3. Energy, Gradient and Hessian as Functions of p

We decompose the energy into likelihood and prior terms:

$$E(p) = M(p) + P(p),$$

with

$$M(p) = \frac{\delta(p)^2}{V(p)},$$

$$P(p) = (p - \mu) (N_{\text{ch}} \Sigma_p^{-1}) (p - \mu)^\top.$$

Here

$$\delta(p) = y - N_{\text{ch}}(p \cdot \gamma), \quad V(p) = \epsilon^2 + N_{\text{ch}}(p \cdot \sigma^2).$$

The full gradient and Hessian, derived in detail in Section 7, can be compactly written as:

$$\mathbf{g}(p) = 2N_{\text{ch}}(p - \mu)\Sigma_p^{-1} - N_{\text{ch}} \left[\frac{2\delta(p)}{V(p)} \gamma^\top + \frac{\delta(p)^2}{V(p)^2} (\sigma^2)^\top \right],$$

and

$$\mathbf{H}(p) = 2N_{\text{ch}} \left[\Sigma_p^{-1} + \frac{N_{\text{ch}}}{V(p)} v(p) v(p)^\top \right],$$

where

$$v(p) = \gamma + \frac{\delta(p)}{V(p)} \sigma^2.$$

These expressions are valid for all p on the simplex. They show explicitly that δ , V and v are functions of p .

4. Rank-1 Structure and Closed-Form Covariance

The key structural observation is that the Hessian is a rank-1 update of the prior precision:

$$\mathbf{H}(p) = 2N_{\text{ch}} \left[\Sigma_p^{-1} + \frac{N_{\text{ch}}}{V(p)} v(p) v(p)^\top \right].$$

Define

$$u(p) = \sqrt{\frac{N_{\text{ch}}}{V(p)}} v(p), \quad \Rightarrow \quad u(p) u(p)^\top = \frac{N_{\text{ch}}}{V(p)} v(p) v(p)^\top.$$

Then

$$\mathbf{H}(p) = 2N_{\text{ch}} (\Sigma_p^{-1} + u(p) u(p)^\top).$$

4.1 Sherman–Morrison Formula

Formally, for an invertible matrix A and a rank-1 update $uu^\top$, the Sherman–Morrison formula says

$$(A + uu^\top)^{-1} = A^{-1} - \frac{A^{-1}uu^\top A^{-1}}{1 + u^\top A^{-1}u}.$$

We apply this with $A = \Sigma_p^{-1}$ and $u = u(p_0)$ at the expansion point p_0 . On the tangent subspace of the simplex, Σ_p has an inverse; at the algebraic level we write $A^{-1} = \Sigma_p$ and obtain

$$(\Sigma_p^{-1} + uu^\top)^{-1} = \Sigma_p - \frac{\Sigma_p uu^\top \Sigma_p}{1 + u^\top \Sigma_p u}.$$

Since

$$\mathbf{H}_0 = 2N_{\text{ch}}(\Sigma_p^{-1} + uu^\top),$$

we get

$$\mathbf{H}_0^{-1} = \frac{1}{2N_{\text{ch}}} \left[\Sigma_p - \frac{\Sigma_p uu^\top \Sigma_p}{1 + u^\top \Sigma_p u} \right].$$

The Laplace covariance at p_0 is

$$\Sigma_{\text{post}} = 2\mathbf{H}_0^{-1} = \frac{1}{N_{\text{ch}}} \left[\Sigma_p - \frac{\Sigma_p uu^\top \Sigma_p}{1 + u^\top \Sigma_p u} \right].$$

4.2 Expressing Everything in Terms of $v(p_0)$

At p_0 we have $u = \sqrt{\frac{N_{\text{ch}}}{V}} v$ with $V = V(p_0)$ and $v = v(p_0)$, so:

$$uu^\top = \frac{N_{\text{ch}}}{V} vv^\top,$$

and we define

$$s = v^\top \Sigma_p v.$$

Then:

$$\begin{aligned} \Sigma_p uu^\top \Sigma_p &= \Sigma_p \left(\frac{N_{\text{ch}}}{V} vv^\top \right) \Sigma_p = \frac{N_{\text{ch}}}{V} \Sigma_p vv^\top \Sigma_p, \\ 1 + u^\top \Sigma_p u &= 1 + \frac{N_{\text{ch}}}{V} v^\top \Sigma_p v = 1 + \frac{N_{\text{ch}}}{V} s = \frac{V + N_{\text{ch}} s}{V}. \end{aligned}$$

Substituting:

$$\Sigma_{\text{post}} = \frac{1}{N_{\text{ch}}} \left[\Sigma_p - \frac{\frac{N_{\text{ch}}}{V} \Sigma_p v v^\top \Sigma_p}{\frac{V + N_{\text{ch}} s}{V}} \right] = \frac{1}{N_{\text{ch}}} \left[\Sigma_p - \frac{N_{\text{ch}}}{V + N_{\text{ch}} s} \Sigma_p v v^\top \Sigma_p \right].$$

Thus we obtain the closed-form Laplace covariance:

$$\Sigma_{\text{post}}(p_0) = \frac{1}{N_{\text{ch}}} \Sigma_p - \frac{1}{V(p_0) + N_{\text{ch}} s(p_0)} \Sigma_p v(p_0) v(p_0)^\top \Sigma_p$$

with $s(p_0) = v(p_0)^\top \Sigma_p v(p_0)$.

Singular covariance and implementation. Note that:

- The derivation uses Σ_p^{-1} only as a formal device on the intrinsic $(d-1)$ -dimensional subspace.
- The final expression for Σ_{post} involves only Σ_p , v and s ; no explicit inverse of Σ_p appears.
- This formula remains valid even when Σ_p is singular in ambient coordinates, so it is suitable for direct implementation.

5. Mean Update and Elimination of Σ_p^{-1}

The Laplace mean at p_0 is

$$p_{\text{post}} \approx p_0 - \mathbf{g}_0 \mathbf{H}_0^{-1} = p_0 - \frac{1}{2} \mathbf{g}_0 \Sigma_{\text{post}},$$

with

$$\mathbf{g}_0 = 2N_{\text{ch}}(p_0 - \mu) \Sigma_p^{-1} - N_{\text{ch}} \left[\frac{2\delta}{V} \gamma^\top + \frac{\delta^2}{V^2} (\sigma^2)^\top \right],$$

where $\delta = \delta(p_0)$ and $V = V(p_0)$.

Introduce

$$\Delta p = p_0 - \mu,$$

and rewrite:

$$\mathbf{g}_0 = 2N_{\text{ch}} \Delta p \Sigma_p^{-1} - N_{\text{ch}} \left[\frac{2\delta}{V} \gamma^\top + \frac{\delta^2}{V^2} (\sigma^2)^\top \right].$$

Then

$$p_{\text{post}} = p_0 - \frac{1}{2} \mathbf{g}_0^\top \Sigma_{\text{post}} = p_0 - N_{\text{ch}} \Delta p \Sigma_p^{-1} \Sigma_{\text{post}} + \frac{N_{\text{ch}}}{2} \left[\frac{2\delta}{V} \gamma^\top + \frac{\delta^2}{V^2} (\sigma^2)^\top \right] \Sigma_{\text{post}}.$$

Equivalently,

$$p_{\text{post}} = p_0 - N_{\text{ch}} \Delta p \Sigma_p^{-1} \Sigma_{\text{post}} + \frac{N_{\text{ch}} \delta}{V} \gamma^\top \Sigma_{\text{post}} + \frac{N_{\text{ch}} \delta^2}{2V^2} (\sigma^2)^\top \Sigma_{\text{post}}.$$

5.1 Eliminating Σ_p^{-1} Using Σ_{post}

We now use the explicit expression for

$$\Sigma_{\text{post}} = \frac{1}{N_{\text{ch}}} \Sigma_p - \frac{1}{V + N_{\text{ch}} s} \Sigma_p v v^\top \Sigma_p$$

to compute $\Sigma_p^{-1} \Sigma_{\text{post}}$ and thereby eliminate Σ_p^{-1} from the mean.

For notational compactness in this subsection, write $N = N_{\text{ch}}$, remembering they are the same quantity.

First:

$$\begin{aligned} \Sigma_p^{-1} \Sigma_{\text{post}} &= \Sigma_p^{-1} \left(\frac{1}{N} \Sigma_p - \frac{1}{V + N s} \Sigma_p v v^\top \Sigma_p \right) \\ &= \frac{1}{N} \underbrace{\Sigma_p^{-1} \Sigma_p}_I - \frac{1}{V + N s} v v^\top \Sigma_p \\ &= \frac{1}{N} I - \frac{1}{V + N s} v v^\top \Sigma_p. \end{aligned}$$

Therefore:

$$N \Delta p \Sigma_p^{-1} \Sigma_{\text{post}} = \Delta p - \frac{N}{V + N s} \Delta p v v^\top \Sigma_p.$$

Substituting into the expression for p_{post} :

$$\begin{aligned} p_{\text{post}} &= p_0 - \left[\Delta p - \frac{N}{V + N s} \Delta p v v^\top \Sigma_p \right] + \frac{N \delta}{V} \gamma^\top \Sigma_{\text{post}} + \frac{N \delta^2}{2V^2} (\sigma^2)^\top \Sigma_{\text{post}} \\ &= (p_0 - \Delta p) + \frac{N}{V + N s} \Delta p v v^\top \Sigma_p + \frac{N \delta}{V} \gamma^\top \Sigma_{\text{post}} + \frac{N \delta^2}{2V^2} (\sigma^2)^\top \Sigma_{\text{post}}. \end{aligned}$$

Since $p_0 - \Delta p = \mu$, we obtain:

$$\begin{aligned} p_{\text{post}} &= \mu + \frac{N_{\text{ch}}}{V + N_{\text{ch}} s} (p_0 - \mu) v v^\top \Sigma_p \\ &\quad + \frac{N_{\text{ch}} \delta}{V} \gamma^\top \Sigma_{\text{post}} + \frac{N_{\text{ch}} \delta^2}{2V^2} (\sigma^2)^\top \Sigma_{\text{post}}. \end{aligned}$$

(2)

Here Σ_{post} is already expressed purely in terms of Σ_p , v , s and V . No explicit Σ_p^{-1} survives: it has been completely cancelled.

Interpretation.

- The first line in (2) pulls the mean back towards the prior mean μ , modulated by the rank-1 direction $vv^\top \Sigma_p$ and the competition between V and $N_{\text{ch}}s$.
- The second line captures the direct influence of the residual δ through the conductance and intrinsic variance vectors (γ, σ^2) , weighted by the covariance Σ_{post} .
- Equation (2) is the general mean update that should be implemented numerically; it only requires multiplications by Σ_p and Σ_{post} , never their inverses.

5.2 Special Case: Expansion at the Prior Mean $p_0 = \mu$

Choosing $p_0 = \mu$ simplifies the expression significantly:

- $\Delta p = p_0 - \mu = 0$,
- the first line in (2) vanishes,
- the gradient at μ reduces to

$$\mathbf{g}_0 = -N_{\text{ch}} \frac{\delta}{V} (\gamma^\top + v^\top),$$

so

$$p_{\text{post}} = \mu - \frac{1}{2} \mathbf{g}_0 \Sigma_{\text{post}} = \mu + \frac{N_{\text{ch}} \delta}{2V} (\gamma^\top + v^\top) \Sigma_{\text{post}}.$$

In this special case we obtain the concise one-step update:

$$p_{\text{post}} \approx \mu + \frac{N_{\text{ch}} \delta}{2V} (\gamma^\top + v^\top) \Sigma_{\text{post}},$$

with

- $\delta = y - N_{\text{ch}}(\mu \cdot \gamma)$,
- $V = \epsilon^2 + N_{\text{ch}}(\mu \cdot \sigma^2)$,
- $v = \gamma + \frac{\delta}{V} \sigma^2$,

- Σ_{post} as in Section 4 with $p_0 = \mu$.

This corresponds to a single Newton step from the prior mean, with a Laplace covariance frozen at that expansion point.

6. Newton–MAP Scheme and MAP–Laplace Posterior

Up to now, p_0 has been a generic expansion point. We now make explicit that:

- the posterior mean of interest is the *MAP* $p^\star$, i.e. the minimum of $E(p)$;
- the covariance we associate with this mean is the *Laplace covariance* Σ_{Lap} given by the curvature at $p^\star$.

6.1 Defining the MAP

The posterior mode (MAP) is defined by

$$p^\star = \arg \min_p E(p)$$

subject to the simplex constraints $p_i \geq 0$ and $\sum_i p_i = 1$. We seek this mode via Newton’s method applied to $E(p)$.

6.2 Newton Iteration

Let $p^{(k)}$ denote the current iterate (row vector). One Newton step is:

$$p^{(k+1)} = p^{(k)} - \Delta p^{(k)}, \quad \mathbf{H}(p^{(k)}) \Delta p^{(k)\top} = \mathbf{g}(p^{(k)})^\top.$$

Algorithmically:

1. Initialise with a feasible point, typically $p^{(0)} = \mu$.
2. For $k = 0, 1, 2, \dots$ until convergence:
 - (a) Compute $\delta^{(k)} = \delta(p^{(k)})$, $V^{(k)} = V(p^{(k)})$, $v^{(k)} = v(p^{(k)})$.
 - (b) Form $\mathbf{g}^{(k)} = \mathbf{g}(p^{(k)})$ and $\mathbf{H}^{(k)} = \mathbf{H}(p^{(k)})$ using the formulas in Section 3.

- (c) Compute the Laplace covariance $\Sigma_{\text{post}}^{(k)}$ at $p^{(k)}$ using the closed form in Section 4.
- (d) Update the mean using the general formula (2) with $p_0 = p^{(k)}$ and $\Sigma_{\text{post}} = \Sigma_{\text{post}}^{(k)}$:

$$p^{(k+1)} = \mu + \frac{N_{\text{ch}}}{V^{(k)} + N_{\text{ch}} s^{(k)}} (p^{(k)} - \mu) v^{(k)} v^{(k)\top} \Sigma_p + \frac{N_{\text{ch}} \delta^{(k)}}{V^{(k)}} \gamma^\top \Sigma_{\text{post}}^{(k)} + \frac{N_{\text{ch}} \delta^{(k)2}}{2V^{(k)2}} (\sigma^2)^\top \Sigma_{\text{post}}^{(k)}.$$

Here $s^{(k)} = v^{(k)\top} \Sigma_p v^{(k)}$.

- (e) Optionally project $p^{(k+1)}$ back onto the simplex (enforce $p_i^{(k+1)} \geq 0$ and renormalise $\sum_i p_i^{(k+1)} = 1$).

If we stop after a single iteration from $p^{(0)} = \mu$, we recover the one-step expression of Section 5.2. In general, Newton generates a sequence $p^{(k)}$ converging to the MAP p^* , and we then set:

$$p_{\text{MAP}} = p^*.$$

6.3 MAP–Laplace Posterior

Once Newton has converged to p^* , we define the *MAP–Laplace posterior* as:

$$p \mid y \approx \mathcal{N}(p^*, \Sigma_{\text{Lap}}),$$

with covariance given by the inverse curvature at p^* :

$$\Sigma_{\text{Lap}} = \mathbf{H}(p^*)^{-1}.$$

Using the rank-1 structure, we compute Σ_{Lap} via the same closed form as $\Sigma_{\text{post}}(p_0)$, simply evaluated at $p_0 = p^*$:

$$\Sigma_{\text{Lap}} = \frac{1}{N_{\text{ch}}} \Sigma_p - \frac{1}{V(p^*) + N_{\text{ch}} s(p^*)} \Sigma_p v(p^*) v(p^*)^\top \Sigma_p,$$

with $s(p^*) = v(p^*)^\top \Sigma_p v(p^*)$.

Thus the computational recipe is:

- iterate Newton using (2) until convergence to p^* ;
- evaluate the rank-1 covariance formula at p^* to obtain Σ_{Lap} .

7. Detailed Derivation of Gradient and Hessian

For completeness, we include the full derivation of the gradient and Hessian of $E(p)$.

Recall:

$$E(p) = M(p) + P(p), \quad M(p) = \frac{\delta^2}{V}, \quad P(p) = (p - \mu) (N_{\text{ch}} \Sigma_p^{-1}) (p - \mu)^\top,$$

with

$$\delta = y - N_{\text{ch}}(p \cdot \gamma), \quad V = \epsilon^2 + N_{\text{ch}}(p \cdot \sigma^2).$$

Here p is a row vector, γ and σ^2 are column vectors.

7.1 Prior Gradient and Hessian

Define the precision matrix $\Lambda_p = N_{\text{ch}} \Sigma_p^{-1}$. Then:

$$P(p) = (p - \mu) \Lambda_p (p - \mu)^\top.$$

Using the standard formula for the gradient of a quadratic form $f(p) = (p - \mu) A (p - \mu)^\top$, with symmetric A , we have:

$$\nabla f(p) = 2(p - \mu) A,$$

and

$$\nabla^2 f(p) = 2A.$$

Thus:

$$\begin{aligned} \nabla P(p) &= 2(p - \mu) \Lambda_p = 2N_{\text{ch}}(p - \mu) \Sigma_p^{-1}, \\ \nabla^2 P(p) &= 2\Lambda_p = 2N_{\text{ch}} \Sigma_p^{-1}. \end{aligned}$$

7.2 Likelihood Gradient

We have:

$$M(p) = \frac{\delta^2}{V}, \quad \delta = y - N_{\text{ch}}(p \cdot \gamma), \quad V = \epsilon^2 + N_{\text{ch}}(p \cdot \sigma^2).$$

First compute the gradients of δ and V :

$$\nabla \delta(p) = -N_{\text{ch}} \gamma^\top, \quad \nabla V(p) = N_{\text{ch}}(\sigma^2)^\top,$$

both row vectors.

Now apply the quotient rule to $M(p) = \delta^2/V$:

$$\nabla M(p) = \frac{2\delta}{V} \nabla \delta - \frac{\delta^2}{V^2} \nabla V.$$

Substituting:

$$\nabla M(p) = \frac{2\delta}{V} (-N_{\text{ch}} \gamma^\top) - \frac{\delta^2}{V^2} N_{\text{ch}} (\sigma^2)^\top = -N_{\text{ch}} \left[\frac{2\delta}{V} \gamma^\top + \frac{\delta^2}{V^2} (\sigma^2)^\top \right].$$

7.3 Likelihood Hessian

Write:

$$M(p) = \delta^2 V^{-1}.$$

We already have:

$$\nabla M(p) = a(p) \nabla \delta(p) - b(p) \nabla V(p),$$

with:

$$a(p) = \frac{2\delta}{V}, \quad b(p) = \frac{\delta^2}{V^2}.$$

Because δ and V are affine in p , their Hessians vanish: $\nabla^2 \delta = 0$, $\nabla^2 V = 0$. Thus:

$$\nabla^2 M(p) = (\nabla a) \nabla \delta - (\nabla b) \nabla V.$$

It remains to compute ∇a and ∇b .

Gradient of $a(p)$.

$$a(p) = \frac{2\delta}{V}.$$

Using the quotient rule:

$$\nabla a = \frac{2}{V} \nabla \delta - \frac{2\delta}{V^2} \nabla V.$$

Substitute:

$$\nabla a = \frac{2}{V} (-N_{\text{ch}} \gamma^\top) - \frac{2\delta}{V^2} N_{\text{ch}} (\sigma^2)^\top = -\frac{2N_{\text{ch}}}{V} \gamma^\top - \frac{2N_{\text{ch}}\delta}{V^2} (\sigma^2)^\top.$$

Gradient of $b(p)$.

$$b(p) = \frac{\delta^2}{V^2}.$$

Differentiating:

$$\nabla b = 2 \frac{\delta}{V^2} \nabla \delta - 2 \frac{\delta^2}{V^3} \nabla V.$$

Substitute:

$$\nabla b = 2 \frac{\delta}{V^2} (-N_{\text{ch}} \gamma^\top) - 2 \frac{\delta^2}{V^3} N_{\text{ch}} (\sigma^2)^\top = -\frac{2N_{\text{ch}}\delta}{V^2} \gamma^\top - \frac{2N_{\text{ch}}\delta^2}{V^3} (\sigma^2)^\top.$$

Assembling the Hessian. Recall:

$$\nabla^2 M(p) = (\nabla a) \nabla \delta - (\nabla b) \nabla V.$$

Using:

$$\nabla \delta = -N_{\text{ch}} \gamma^\top, \quad \nabla V = N_{\text{ch}} (\sigma^2)^\top,$$

we obtain:

$$\nabla^2 M(p) = (\nabla a) (-N_{\text{ch}} \gamma^\top) - (\nabla b) N_{\text{ch}} (\sigma^2)^\top.$$

Substituting the expressions for ∇a and ∇b and collecting terms, we arrive at:

$$\nabla^2 M(p) = 2N_{\text{ch}}^2 \left[\frac{1}{V} \gamma \gamma^\top + \frac{\delta}{V^2} (\gamma (\sigma^2)^\top + \sigma^2 \gamma^\top) + \frac{\delta^2}{V^3} \sigma^2 (\sigma^2)^\top \right].$$

7.4 Full Gradient and Hessian

Combining prior and likelihood:

$$\begin{aligned} \mathbf{g}(p) &= \nabla E(p) = \nabla M(p) + \nabla P(p) \\ &= 2N_{\text{ch}}(p - \mu) \Sigma_p^{-1} - N_{\text{ch}} \left[\frac{2\delta}{V} \gamma^\top + \frac{\delta^2}{V^2} (\sigma^2)^\top \right], \end{aligned}$$

and:

$$\begin{aligned} \mathbf{H}(p) &= \nabla^2 E(p) = \nabla^2 M(p) + \nabla^2 P(p) \\ &= 2N_{\text{ch}} \Sigma_p^{-1} + 2N_{\text{ch}}^2 \left[\frac{1}{V} \gamma \gamma^\top + \frac{\delta}{V^2} (\gamma (\sigma^2)^\top + \sigma^2 \gamma^\top) + \frac{\delta^2}{V^3} \sigma^2 (\sigma^2)^\top \right]. \end{aligned}$$

7.5 Rank-1 Factorisation of the Hessian

Define:

$$v = \gamma + \frac{\delta}{V}\sigma^2.$$

Compute:

$$vv^\top = \gamma\gamma^\top + \frac{\delta}{V}(\gamma(\sigma^2)^\top + \sigma^2\gamma^\top) + \frac{\delta^2}{V^2}\sigma^2(\sigma^2)^\top.$$

Then:

$$\frac{N_{\text{ch}}}{V}vv^\top = \frac{N_{\text{ch}}}{V}\gamma\gamma^\top + \frac{N_{\text{ch}}\delta}{V^2}(\gamma(\sigma^2)^\top + \sigma^2\gamma^\top) + \frac{N_{\text{ch}}\delta^2}{V^3}\sigma^2(\sigma^2)^\top,$$

which matches exactly the bracketed term in the expression for $\nabla^2 M(p)$, scaled by $2N_{\text{ch}}^2$.

Thus:

$$\nabla^2 M(p) = 2N_{\text{ch}}^2 \frac{1}{V}vv^\top,$$

and so the full Hessian is:

$$\mathbf{H}(p) = 2N_{\text{ch}}\Sigma_p^{-1} + 2N_{\text{ch}}^2 \frac{1}{V}vv^\top = 2N_{\text{ch}} \left[\Sigma_p^{-1} + \frac{N_{\text{ch}}}{V}vv^\top \right].$$

Introducing

$$u = \sqrt{\frac{N_{\text{ch}}}{V}} v,$$

we recover:

$$\mathbf{H}(p) = 2N_{\text{ch}}(\Sigma_p^{-1} + uu^\top),$$

which is the starting point for the Sherman–Morrison inversion in Section 4.

This completes the detailed derivation with consistent vector orientations, explicit dependence of δ , V and v on p , and with all explicit uses of Σ_p^{-1} eliminated from the implementable formulas for p_{post} and Σ_{post} .

8. Exact MAP Energy with $\log V$ and Rank-2 Hessian

In the previous sections we defined the energy as

$$E_{\text{quasi}}(p) = \frac{\delta(p)^2}{V(p)} + (p - \mu)(N_{\text{ch}}\Sigma_p^{-1})(p - \mu)^\top,$$

which corresponds to a *quasi-likelihood* that uses $V(p)$ only as a weight for the squared residual. This choice yields a rank-1 Hessian update and a simple closed form for Σ_{post} .

If we take seriously the generative model

$$y \mid p \sim \mathcal{N}(N_{\text{ch}}(p \cdot \gamma), V(p)),$$

then the *exact* negative log-likelihood (up to an additive constant) is

$$E_{\text{like}}(p) = \log V(p) + \frac{\delta(p)^2}{V(p)}.$$

In this section we incorporate the $\log V(p)$ term and work with the *exact* negative log-posterior

$$E_{\text{exact}}(p) = \underbrace{\log V(p) + \frac{\delta(p)^2}{V(p)}}_{\text{data}} + \underbrace{(p - \mu)(N_{\text{ch}}\Sigma_p^{-1})(p - \mu)^\top}_{\text{prior}} + \text{const.}$$

As before:

$$\delta(p) = y - N_{\text{ch}}(p \cdot \gamma), \quad V(p) = \epsilon^2 + N_{\text{ch}}(p \cdot \sigma^2),$$

and

$$v(p) = \gamma + \frac{\delta(p)}{V(p)} \sigma^2.$$

The prior log-determinant term (pseudo-determinant of the singular Σ_p) is a constant in p and therefore *does not* affect the MAP; we ignore it here.

8.1 Exact Gradient

We decompose

$$E_{\text{exact}}(p) = L(p) + M(p) + P(p),$$

with

$$L(p) = \log V(p), \quad M(p) = \frac{\delta(p)^2}{V(p)}, \quad P(p) = (p - \mu)(N_{\text{ch}}\Sigma_p^{-1})(p - \mu)^\top.$$

The gradients of M and P were already computed in Section 7:

$$\nabla M(p) = -N_{\text{ch}} \left[\frac{2\delta(p)}{V(p)} \gamma^\top + \frac{\delta(p)^2}{V(p)^2} (\sigma^2)^\top \right],$$

$$\nabla P(p) = 2N_{\text{ch}}(p - \mu)\Sigma_p^{-1}.$$

For the new term $L(p) = \log V(p)$ we have

$$\nabla L(p) = \frac{1}{V(p)} \nabla V(p) = \frac{N_{\text{ch}}}{V(p)} (\sigma^2)^\top,$$

since $\nabla V(p) = N_{\text{ch}}(\sigma^2)^\top$.

Thus the *exact* gradient becomes

$$\mathbf{g}_{\text{exact}}(p) = \nabla E_{\text{exact}}(p) = 2N_{\text{ch}}(p - \mu)\Sigma_p^{-1} - N_{\text{ch}} \left[\frac{2\delta(p)}{V(p)} \gamma^\top + \frac{\delta(p)^2}{V(p)^2} (\sigma^2)^\top - \frac{1}{V(p)} (\sigma^2)^\top \right].$$

Equivalently, grouping the $(\sigma^2)^\top$ terms:

$$\boxed{\mathbf{g}_{\text{exact}}(p) = 2N_{\text{ch}}(p - \mu)\Sigma_p^{-1} - N_{\text{ch}} \left[\frac{2\delta(p)}{V(p)} \gamma^\top + \left(\frac{\delta(p)^2}{V(p)^2} - \frac{1}{V(p)} \right) (\sigma^2)^\top \right].}$$

This is what must be used in a true Newton iteration targeting the exact MAP.

8.2 Exact Hessian and Rank-2 Structure

We now add the Hessian of $L(p)$ to the Hessian derived in Section 7 for $M(p) + P(p)$.

From Section 7:

$$\nabla^2 P(p) = 2N_{\text{ch}}\Sigma_p^{-1},$$

$$\nabla^2 M(p) = 2N_{\text{ch}}^2 \left[\frac{1}{V(p)} \gamma\gamma^\top + \frac{\delta(p)}{V(p)^2} (\gamma(\sigma^2)^\top + \sigma^2\gamma^\top) + \frac{\delta(p)^2}{V(p)^3} \sigma^2\sigma^{2\top} \right].$$

For $L(p) = \log V(p)$, using $\nabla V(p) = N_{\text{ch}}(\sigma^2)^\top$ and $\nabla^2 V(p) = 0$:

$$\nabla^2 L(p) = -\frac{1}{V(p)^2} (\nabla V(p))^\top \nabla V(p) = -\frac{N_{\text{ch}}^2}{V(p)^2} \sigma^2\sigma^{2\top}.$$

Therefore:

$$\nabla^2 (M(p) + L(p)) = 2N_{\text{ch}}^2 \left[\frac{1}{V(p)} \gamma\gamma^\top + \frac{\delta(p)}{V(p)^2} (\gamma(\sigma^2)^\top + \sigma^2\gamma^\top) + \left(\frac{\delta(p)^2}{V(p)^3} - \frac{1}{2V(p)^2} \right) \sigma^2\sigma^{2\top} \right].$$

Adding the prior Hessian,

$$\nabla^2 P(p) = 2N_{\text{ch}}\Sigma_p^{-1},$$

we obtain the full exact Hessian:

$$\mathbf{H}_{\text{exact}}(p) = 2N_{\text{ch}}\Sigma_p^{-1} + 2N_{\text{ch}}^2 \left[\frac{1}{V(p)} \gamma \gamma^\top + \frac{\delta(p)}{V(p)^2} (\gamma(\sigma^2)^\top + \sigma^2 \gamma^\top) + \left(\frac{\delta(p)^2}{V(p)^3} - \frac{1}{2V(p)^2} \right) \sigma^2 \sigma^{2\top} \right].$$

It is convenient to rewrite this as a *rank-2* update of the prior precision. As before, define

$$v(p) = \gamma + \frac{\delta(p)}{V(p)} \sigma^2,$$

and let

$$u_1(p) = v(p), \quad u_2(p) = \sigma^2.$$

We will write the Hessian in the form

$$\mathbf{H}_{\text{exact}}(p) = 2N_{\text{ch}} [\Sigma_p^{-1} + U(p)C(p)U(p)^\top],$$

with

$$U(p) = [u_1(p), u_2(p)] \in \mathbb{R}^{d \times 2},$$

and $C(p)$ a 2×2 diagonal matrix.

From the rank-1 case we know that

$$\frac{N_{\text{ch}}}{V(p)} v(p)v(p)^\top = \frac{N_{\text{ch}}}{V(p)} u_1(p)u_1(p)^\top$$

generates the M -part of the Hessian without $\log V$. The extra term from $L(p)$ is

$$-\frac{N_{\text{ch}}^2}{V(p)^2} \sigma^2 \sigma^{2\top} = 2N_{\text{ch}} \left(-\frac{N_{\text{ch}}}{2V(p)^2} \right) u_2(p)u_2(p)^\top.$$

Thus we can take

$$C(p) = \begin{pmatrix} \alpha(p) & 0 \\ 0 & \beta(p) \end{pmatrix}, \quad \alpha(p) = \frac{N_{\text{ch}}}{V(p)}, \quad \beta(p) = -\frac{N_{\text{ch}}}{2V(p)^2},$$

so that

$$U(p)C(p)U(p)^\top = \alpha(p)u_1(p)u_1(p)^\top + \beta(p)u_2(p)u_2(p)^\top.$$

We have therefore

$$\boxed{\mathbf{H}_{\text{exact}}(p) = 2N_{\text{ch}} [\Sigma_p^{-1} + U(p)C(p)U(p)^\top]},$$

with a *rank-2* update governed by $v(p)$ and σ^2 .

8.3 Woodbury Inversion and Exact Laplace Covariance

The Woodbury identity for a rank- k update says that, for invertible A ,

$$(A + UCU^\top)^{-1} = A^{-1} - A^{-1}U(C^{-1} + U^\top A^{-1}U)^{-1}U^\top A^{-1},$$

where U is $d \times k$ and C is $k \times k$.

We apply this in the intrinsic $(d-1)$ -dimensional subspace with

$$A = \Sigma_p^{-1}, \quad A^{-1} = \Sigma_p, \quad U = U(p_0), \quad C = C(p_0),$$

for an expansion point p_0 . Then

$$(\Sigma_p^{-1} + UCU^\top)^{-1} = \Sigma_p - \Sigma_p U K U^\top \Sigma_p,$$

where

$$K = (C^{-1} + U^\top \Sigma_p U)^{-1} \in \mathbb{R}^{2 \times 2}.$$

Since

$$\mathbf{H}_{\text{exact}}(p_0) = 2N_{\text{ch}}(\Sigma_p^{-1} + UCU^\top),$$

we get

$$\mathbf{H}_{\text{exact}}(p_0)^{-1} = \frac{1}{2N_{\text{ch}}} \left[\Sigma_p - \Sigma_p U K U^\top \Sigma_p \right].$$

The *exact* Laplace covariance at p_0 is

$$\Sigma_{\text{post}}^{\text{exact}}(p_0) = 2\mathbf{H}_{\text{exact}}(p_0)^{-1} = \frac{1}{N_{\text{ch}}} \left[\Sigma_p - \Sigma_p U K U^\top \Sigma_p \right].$$

Explicitly, at p_0 :

$$U = [v(p_0), \sigma^2], \quad C = \begin{pmatrix} \alpha & 0 \\ 0 & \beta \end{pmatrix},$$

with

$$\alpha = \frac{N_{\text{ch}}}{V(p_0)}, \quad \beta = -\frac{N_{\text{ch}}}{2V(p_0)^2}.$$

The 2×2 matrix

$$M = C^{-1} + U^\top \Sigma_p U$$

is

$$M = \begin{pmatrix} \alpha^{-1} + v^\top \Sigma_p v & v^\top \Sigma_p \sigma^2 \\ \sigma^{2\top} \Sigma_p v & \beta^{-1} + \sigma^{2\top} \Sigma_p \sigma^2 \end{pmatrix} = \begin{pmatrix} \frac{V}{N_{\text{ch}}} + s & b \\ b & -\frac{2V(p_0)^2}{N_{\text{ch}}} + c \end{pmatrix},$$

where

$$s = v^\top \Sigma_p v, \quad b = v^\top \Sigma_p \sigma^2, \quad c = \sigma^2 \Sigma_p \sigma^2.$$

Then

$$K = M^{-1}$$

is the inverse of a *2-by-2* matrix, which can always be computed analytically or numerically with negligible cost relative to the channel-ensemble computations.

Thus the exact MAP covariance is

$$\Sigma_{\text{post}}^{\text{exact}}(p_0) = \frac{1}{N_{\text{ch}}} \left[\Sigma_p - \Sigma_p U(p_0) K(p_0) U(p_0)^\top \Sigma_p \right],$$

with $K(p_0) = (C(p_0)^{-1} + U(p_0)^\top \Sigma_p U(p_0))^{-1}$.

Remark (singular prior). Again, Σ_p^{-1} appears only formally in the Woodbury algebra. The implementable expression for $\Sigma_{\text{post}}^{\text{exact}}(p_0)$ contains only Σ_p , $U(p_0)$ and the small 2×2 inverse $K(p_0)$.

8.4 Exact Mean Update and Elimination of Σ_p^{-1}

The exact Laplace mean at p_0 is

$$p_{\text{post}}^{\text{exact}} = p_0 - \mathbf{g}_{\text{exact}}(p_0) \mathbf{H}_{\text{exact}}(p_0)^{-1} = p_0 - \frac{1}{2} \mathbf{g}_{\text{exact}}(p_0) \Sigma_{\text{post}}^{\text{exact}}(p_0).$$

Write

$$\Delta p = p_0 - \mu,$$

and decompose the gradient as

$$\mathbf{g}_{\text{exact}}(p_0) = 2N_{\text{ch}} \Delta p \Sigma_p^{-1} - N_{\text{ch}} q^\top,$$

where

$$q^\top = \frac{2\delta}{V} \gamma^\top + \left(\frac{\delta^2}{V^2} - \frac{1}{V} \right) (\sigma^2)^\top,$$

with $\delta = \delta(p_0)$ and $V = V(p_0)$.

Then

$$p_{\text{post}}^{\text{exact}} = p_0 - N_{\text{ch}} \Delta p \Sigma_p^{-1} \Sigma_{\text{post}}^{\text{exact}}(p_0) + \frac{N_{\text{ch}}}{2} q^\top \Sigma_{\text{post}}^{\text{exact}}(p_0).$$

Using the exact covariance expression

$$\Sigma_{\text{post}}^{\text{exact}}(p_0) = \frac{1}{N_{\text{ch}}} \left[\Sigma_p - \Sigma_p U K U^\top \Sigma_p \right],$$

we compute

$$\begin{aligned} \Sigma_p^{-1} \Sigma_{\text{post}}^{\text{exact}}(p_0) &= \Sigma_p^{-1} \left[\frac{1}{N_{\text{ch}}} (\Sigma_p - \Sigma_p U K U^\top \Sigma_p) \right] \\ &= \frac{1}{N_{\text{ch}}} (I - U K U^\top \Sigma_p). \end{aligned}$$

Thus

$$N_{\text{ch}} \Delta p \Sigma_p^{-1} \Sigma_{\text{post}}^{\text{exact}}(p_0) = \Delta p - \Delta p U K U^\top \Sigma_p.$$

Substituting back:

$$\begin{aligned} p_{\text{post}}^{\text{exact}} &= p_0 - \left[\Delta p - \Delta p U K U^\top \Sigma_p \right] + \frac{N_{\text{ch}}}{2} q^\top \Sigma_{\text{post}}^{\text{exact}}(p_0) \\ &= (p_0 - \Delta p) + \Delta p U K U^\top \Sigma_p + \frac{N_{\text{ch}}}{2} q^\top \Sigma_{\text{post}}^{\text{exact}}(p_0). \end{aligned}$$

Since $p_0 - \Delta p = \mu$, we obtain the exact mean update in a form with no explicit Σ_p^{-1} :

$$\boxed{p_{\text{post}}^{\text{exact}} = \mu + \Delta p U(p_0) K(p_0) U(p_0)^\top \Sigma_p + \frac{N_{\text{ch}}}{2} q^\top \Sigma_{\text{post}}^{\text{exact}}(p_0),}$$

with

$$\Delta p = p_0 - \mu, \quad q^\top = \frac{2\delta}{V} \gamma^\top + \left(\frac{\delta^2}{V^2} - \frac{1}{V} \right) (\sigma^2)^\top,$$

and $\Sigma_{\text{post}}^{\text{exact}}(p_0)$ given above.

8.5 Summary: Rank-1 vs Rank-2 Schemes

We now have two related Laplace schemes:

- **Quasi-Laplace (rank-1):**

- Energy: $E_{\text{quasi}}(p) = \delta(p)^2/V(p) + P(p)$.
- Hessian: rank-1 update, closed form $\Sigma_{\text{post}}(p_0) = \frac{1}{N_{\text{ch}}} \Sigma_p - \frac{1}{V(p_0) + N_{\text{ch}} s(p_0)} \Sigma_p v(p_0) v(p_0)^\top$
- Mean update: Eq. (2) in Section 5, no explicit Σ_p^{-1} .

- **Exact MAP + Laplace (rank-2):**

- Energy: $E_{\text{exact}}(p) = \log V(p) + \delta(p)^2/V(p) + P(p)$.
- Hessian: rank-2 update, $\mathbf{H}_{\text{exact}}(p) = 2N_{\text{ch}}[\Sigma_p^{-1} + U(p)C(p)U(p)^\top]$.
- Covariance: $\Sigma_{\text{post}}^{\text{exact}}(p_0) = \frac{1}{N_{\text{ch}}}[\Sigma_p - \Sigma_p U(p_0)K(p_0)U(p_0)^\top \Sigma_p]$,
with a 2×2 matrix $K(p_0)$.
- Mean update: the exact expression above, again without explicit Σ_p^{-1} .

From an implementation perspective, the heavy lifting in MacroIR/MacroDR remains the computation of the channel-ensemble statistics that define γ , σ^2 , and thus $V(p)$ and $v(p)$. Going from a rank-1 to a rank-2 Hessian only requires a 2×2 matrix inversion per interval, which is negligible compared to the cost of the boundary-state and tilde-operator computations.

8. Exact MAP from the Gaussian Likelihood (with $\frac{1}{2}$ factors)

Up to now we have worked with an “energy”

$$E(p) = M(p) + P(p)$$

(and in Section 8 with $E_{\text{exact}}(p) = L(p) + M(p) + P(p)$) without explicit $\frac{1}{2}$ factors. In this section we start from the *actual* Gaussian likelihood and prior, write the *true* negative log-posterior, and then explain how it relates to the energies used in the rest of the note.

8.1 Gaussian likelihood and prior in standard form

Given a macroscopic state p , the scalar interval-averaged current is modelled as

$$y \mid p \sim \mathcal{N}(\mu_y(p), V(p)), \quad \mu_y(p) = N_{\text{ch}}(p \cdot \gamma),$$

with

$$V(p) = \epsilon^2 + N_{\text{ch}}(p \cdot \sigma^2).$$

The Gaussian likelihood is

$$p(y \mid p) = \frac{1}{\sqrt{2\pi V(p)}} \exp\left(-\frac{1}{2} \frac{(y - \mu_y(p))^2}{V(p)}\right).$$

Define

$$\delta(p) = y - N_{\text{ch}}(p \cdot \gamma),$$

so that

$$p(y | p) = \frac{1}{\sqrt{2\pi V(p)}} \exp\left(-\frac{1}{2} \frac{\delta(p)^2}{V(p)}\right).$$

The log-likelihood is therefore

$$\log p(y | p) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log V(p) - \frac{1}{2} \frac{\delta(p)^2}{V(p)}.$$

Ignoring the constant $-\frac{1}{2} \log(2\pi)$, the *negative* log-likelihood is

$$F_{\text{like}}(p) = \frac{1}{2} \log V(p) + \frac{1}{2} \frac{\delta(p)^2}{V(p)} \quad (+ \text{ const}).$$

For the prior we assume

$$p \sim \mathcal{N}(\mu, \Sigma_p)$$

on the simplex, with precision

$$\Lambda_p = N_{\text{ch}} \Sigma_p^{-1}.$$

The prior density is

$$p(p) \propto \exp\left(-\frac{1}{2} (p - \mu) \Lambda_p (p - \mu)^\top\right),$$

so the negative log-prior (up to a constant) is

$$F_{\text{prior}}(p) = \frac{1}{2} (p - \mu) \Lambda_p (p - \mu)^\top = \frac{1}{2} (p - \mu) (N_{\text{ch}} \Sigma_p^{-1}) (p - \mu)^\top.$$

8.2 Exact negative log-posterior and scaling to “energy”

The (unnormalised) posterior is

$$p(p | y) \propto p(y | p) p(p),$$

so the *exact* negative log-posterior is

$$F_{\text{exact}}(p) = F_{\text{like}}(p) + F_{\text{prior}}(p) \quad (+ \text{ const}),$$

that is

$$F_{\text{exact}}(p) = \frac{1}{2} \log V(p) + \frac{1}{2} \frac{\delta(p)^2}{V(p)} + \frac{1}{2} (p - \mu) (N_{\text{ch}} \Sigma_p^{-1}) (p - \mu)^\top \quad (+ \text{const}).$$

This is what one should minimise to obtain the true MAP:

$$p^\star = \arg \min_p F_{\text{exact}}(p).$$

For algebraic convenience in the rest of the note we work with an *energy* that is simply twice the negative log-posterior:

$$E_{\text{exact}}(p) := 2F_{\text{exact}}(p).$$

Up to an additive constant,

$$E_{\text{exact}}(p) = \underbrace{\log V(p) + \frac{\delta(p)^2}{V(p)}}_{L(p)+M(p)} + \underbrace{(p - \mu) (N_{\text{ch}} \Sigma_p^{-1}) (p - \mu)^\top}_{P(p)}.$$

This is exactly the decomposition used in Section 8:

$$E_{\text{exact}}(p) = L(p) + M(p) + P(p).$$

Important: multiplying F_{exact} by a positive constant does not change the MAP. Moreover, since

$$E_{\text{exact}}(p) = 2F_{\text{exact}}(p),$$

their Hessians satisfy

$$\nabla^2 E_{\text{exact}}(p) = 2 \nabla^2 F_{\text{exact}}(p),$$

and the Laplace covariance of the posterior,

$$\Sigma_{\text{Lap}} \approx [\nabla^2 F_{\text{exact}}(p^\star)]^{-1},$$

can be computed as

$$\Sigma_{\text{Lap}} = 2[\nabla^2 E_{\text{exact}}(p^\star)]^{-1}.$$

This is exactly the convention used earlier in the note, where we wrote

$$\Sigma_{\text{post}} = 2 \mathbf{H}^{-1}$$

with $\mathbf{H} = \nabla^2 E$.

Thus:

- all gradient and Hessian formulas derived for E_{exact} (including the rank-2 Woodbury structure in Section 8) are correct and correspond to $2\times$ the exact negative log-posterior;
- the MAP is obtained by minimising F_{exact} or equivalently E_{exact} ;
- the Laplace covariance is always $\Sigma_{\text{Lap}} = 2\mathbf{H}_{\text{exact}}^{-1}$ when $\mathbf{H}_{\text{exact}} = \nabla^2 E_{\text{exact}}$.

8.3 Exact vs quasi energy

For comparison:

- The *exact* data term from the Gaussian likelihood is

$$F_{\text{like}}(p) = \frac{1}{2} \log V(p) + \frac{1}{2} \frac{\delta(p)^2}{V(p)} \quad (+ \text{const}),$$

and the corresponding energy contribution in this note is

$$L(p) + M(p) = \log V(p) + \frac{\delta(p)^2}{V(p)} = 2F_{\text{like}}(p) \quad (+ \text{const}).$$

- The earlier “quasi” scheme drops the $\log V(p)$ term and keeps only $\delta(p)^2/V(p)$ (again up to a factor 2), which yields a rank-1 Hessian and the simpler covariance formula.

In other words:

- if you want the *exact* Gaussian MAP, work with F_{exact} defined above; all rank-2 Hessian and Woodbury formulas in Section 8 apply directly via $E_{\text{exact}} = 2F_{\text{exact}}$ and $\Sigma_{\text{Lap}} = 2\mathbf{H}_{\text{exact}}^{-1}$;
- if you are happy with the quasi-likelihood (no $\log V$ term), you can use the rank-1 scheme from the earlier sections.